AERO 151 Notice of Intent

Advanced Drone Regulations and Safety Management

Basic Course Information

Subject AERO, Aeronautical and Aviation Technology
Course number 151
Tentative course description This one-unit lecture course develops students' ability to navigate FAA regulatory processes and manage safety in professional drone operations beyond introductory Part 107 preparation. Students examine waivers, airspace authorizations, coordination with aviation authorities, and the supporting documentation needed to explain how a proposed operation can be conducted safely. Topics include operations beyond visual line of sight, Part 108 rulemaking, and other emerging regulatory frameworks, with attention to distinguishing proposed changes from requirements currently in effect. Safety management instruction addresses organizational responsibilities, hazard identification, risk assessment, reporting, and evaluation of whether safety measures are working. Through case studies and scenario exercises, students connect regulatory requirements with practical operating procedures and defensible safety decisions.

Credit and Hours

Variable hours No. Minimum units 1.0. Lecture units 1.0. Lab units 0.
Lecture / lab / activity hours 1 / 0 / 0 curriculum hours per week. Semester totals are 18 lecture, 0 lab, and 0 activity hours. Outside work is 36 hours. Total student hours are 54.

Course Content

Fit within existing degrees or certificates No. Planned certificates of specialization will be proposed separately.
Course rationale Drone regulations continue to evolve, and some enterprise operations require FAA permissions or waivers beyond routine flight operations. Advanced pilots and drone program managers need the ability to evaluate these requirements and manage organizational safety. This course supports professionals responsible for complex operations, including utility work, through advanced regulatory practice and safety management aligned with FAA standards.
Transfer or articulation intent No.
Connection to the College Mission The course supports students' career goals by preparing them for greater responsibility in professional drone programs. It develops skills that can strengthen job performance, career advancement, and earning potential while supporting responsible operations.
Additional resources required _____
Resources and funding _____
Duplicate or overlap existing courses Yes. AERO 100 introduces drone regulations and safety. AERO 151 extends that foundation to advanced permissions, waivers, and organizational safety management. Consultation with related disciplines has established support for the proposed scope.

Minimum Qualifications

Minimum qualification Aeronautics. Condition _____
Department currently has faculty meeting minimum qualifications _____

Course Cap

Tentative course cap category _____ Tentative course cap 25.
Effect on department efficiency A cap of 25 supports individual feedback on regulatory analysis and safety decisions. The smaller section size allows substantive case discussion and review of students' reasoning.

Codes and Dates

Dean or director approval date _____
Originator Kyle Nessen. Co-contributor Ron Overacker.

AERO 161 Notice of Intent

Drone Applications in Geospatial Technology

Basic Course Information

Subject AERO, Aeronautical and Aviation Technology
Course number 161
Tentative course description This course develops the skills needed to collect and prepare reliable spatial data using drones for a broad range of professional applications. Students learn the complete orthomosaic production process, from selecting camera settings and designing flight missions to processing overlapping imagery into a georeferenced image of an area. Field exercises incorporate GNSS receivers, high-accuracy positioning, ground control, and independent checkpoints to establish and evaluate positional accuracy. Instruction also introduces other sensors and data products, allowing students to consider collection methods beyond conventional photography and photogrammetry. A culminating project integrates safe mission planning, field operations, processing, quality checks, and documentation to deliver usable datasets for GIS and other professional applications.

Credit and Hours

Variable hours No. Minimum units 3.0. Lecture units 2.0. Lab units 1.0.
Lecture / lab / activity hours 2 / 3 / 0 curriculum hours per week. Semester totals are 36 lecture, 54 lab, and 0 activity hours. Outside work is 72 hours. Total student hours are 162.

Course Content

Fit within existing degrees or certificates No. Planned certificates of specialization will be proposed separately.
Course rationale Employers need usable, reliable spatial data, and drone operators need more than flight skills to produce it. This course prepares students for mapping work in surveying, construction, environmental monitoring, and infrastructure management. It strengthens employment prospects and helps working professionals add drone data collection to their existing skills. The course will support proposed certificates of specialization.
Transfer or articulation intent No.
Connection to the College Mission The course supports student success and career goals through practical skills in producing reliable spatial data. These skills can improve employment opportunities, job performance, and access to more advanced technical work.
Additional resources required _____
Resources and funding _____
Duplicate or overlap existing courses Yes. Related content appears in GEOL 230, GEOL 233, GEOL 234, and ENGR 201. GIS, engineering, and surveying faculty have been consulted and support the course as a distinct, complementary offering. Its emphasis is drone mission design, aircraft and sensor operation, and production of reliable aerial datasets.

Minimum Qualifications

Minimum qualification Aeronautics. Condition _____
Department currently has faculty meeting minimum qualifications _____

Course Cap

Tentative course cap category 9 - Lab - Small (20-30). Tentative course cap 25.
Effect on department efficiency A cap of 25 matches AERO 100 and supports supervised fieldwork and individual assessment. Small crews allow students to rotate through flight, sensor, and data processing roles while receiving feedback.

Codes and Dates

Dean or director approval date _____
Originator Kyle Nessen. Co-contributor Ron Overacker.

AERO 162 Notice of Intent

Drone Applications in Precision Agriculture

Basic Course Information

Subject AERO, Aeronautical and Aviation Technology
Course number 162
Tentative course description This course introduces drone applications in precision agriculture, using information and technology to support crop management based on conditions at particular locations within a field. Students collect visible and multispectral imagery to document crop variability, support scouting, and identify areas that warrant closer field investigation. Thermal imaging exercises examine temperature patterns associated with crop water stress and irrigation performance, including the conditions that affect interpretation. Agricultural spraying instruction focuses on safely planning and executing drone missions that deliver chemicals to their intended treatment areas through appropriate flight paths, equipment settings, and coverage. The spraying component addresses drone operation and application placement rather than the pesticide applicator knowledge and licensing curriculum taught through specialized agricultural training.

Credit and Hours

Variable hours No. Minimum units 3.0. Lecture units 2.0. Lab units 1.0.
Lecture / lab / activity hours 2 / 3 / 0 curriculum hours per week. Semester totals are 36 lecture, 54 lab, and 0 activity hours. Outside work is 72 hours. Total student hours are 162.

Course Content

Fit within existing degrees or certificates No. Planned certificates of specialization will be proposed separately.
Course rationale Drones are an expanding tool in agriculture, creating opportunities for workers who can apply the technology effectively. This course helps students compete for agricultural jobs and helps working professionals add useful capabilities to their existing roles. Specialized drone skills can support advancement and earning potential in farm operations and agricultural services. The course will support proposed certificates of specialization.
Transfer or articulation intent No.
Connection to the College Mission The course supports students' employment and advancement goals through practical agricultural technology skills. It prepares students to improve their work, pursue new responsibilities, and contribute to the region's agricultural economy.
Additional resources required _____
Resources and funding _____
Duplicate or overlap existing courses Yes. Related concepts appear in AGPS 230, AGPS 242, and GEOL 234. Consultation with related disciplines has established support for the course. Instruction focuses on drone sensing and application operations. Pesticide applicator knowledge and licensing instruction belong in agriculture and are outside this course's scope.

Minimum Qualifications

Minimum qualification Aeronautics. Condition _____
Department currently has faculty meeting minimum qualifications _____

Course Cap

Tentative course cap category 9 - Lab - Small (20-30). Tentative course cap 25.
Effect on department efficiency A cap of 25 matches AERO 100 and supports supervised agricultural flight exercises. Small crews provide opportunities to practice aircraft operation, sensor use, and application placement with individual feedback.

Codes and Dates

Dean or director approval date _____
Originator Kyle Nessen. Co-contributor Ron Overacker.

AERO 163 Notice of Intent

Drone Applications in Public Safety

Basic Course Information

Subject AERO, Aeronautical and Aviation Technology
Course number 163
Tentative course description This course prepares students to support public safety missions with drones in search and rescue, fire response, law enforcement, and emergency management. Students develop skills in visible and thermal imaging to search for missing people, assess incident conditions, and document affected areas. Practical exercises include photogrammetric reconstruction of simulated collision and crime scenes, giving students experience capturing imagery and producing spatial records for investigation and documentation. Additional scenarios develop search planning, disaster assessment, and live aerial observation skills while emphasizing crew coordination, communication with incident command, and safe operation in changing environments. Students examine the legal and regulatory frameworks specific to public safety drone operations, including emergency airspace access, agency operating authority, privacy protections, and the handling of imagery and evidence.

Credit and Hours

Variable hours No. Minimum units 3.0. Lecture units 2.0. Lab units 1.0.
Lecture / lab / activity hours 2 / 3 / 0 curriculum hours per week. Semester totals are 36 lecture, 54 lab, and 0 activity hours. Outside work is 72 hours. Total student hours are 162.

Course Content

Fit within existing degrees or certificates No. Planned certificates of specialization will be proposed separately.
Course rationale Public safety agencies increasingly use drones as a tool for response, assessment, and documentation. This course helps existing public safety professionals develop additional operational capabilities and helps students pursuing those careers build relevant technical skills. It supports improved job performance and opportunities for specialized assignments and advancement. The course will support proposed certificates of specialization and complement preparation for the underlying public safety roles.
Transfer or articulation intent No.
Connection to the College Mission The course advances students' career goals by developing technical skills relevant to public safety employment and professional growth. Students learn to improve operational work while serving community needs and maintaining public trust.
Additional resources required _____
Resources and funding _____
Duplicate or overlap existing courses Yes. Related content appears in CJ 210 and CJ 213, with additional connections to GIS. Consultation with related disciplines has established support for the course. Its distinct focus is drone operation, aerial data collection, and crew coordination in public safety scenarios.

Minimum Qualifications

Minimum qualification Aeronautics. Condition _____
Department currently has faculty meeting minimum qualifications _____

Course Cap

Tentative course cap category 9 - Lab - Small (20-30). Tentative course cap 25.
Effect on department efficiency A cap of 25 matches AERO 100 and supports supervised scenario exercises. Small crews allow individual assessment of flight, imaging, communication, and coordination skills while students rotate through operational roles.

Codes and Dates

Dean or director approval date _____
Originator Kyle Nessen. Co-contributor Ron Overacker.

AERO 147 Notice of Intent

Independent Studies in Drone Technology

Basic Course Information

Subject AERO, Aeronautical and Aviation Technology
Course number 147 (proposed)
Tentative course description This variable-unit course provides prepared students an opportunity for in-depth study or an applied project in drone technology under the direction of an instructor. Students build on completed preparatory coursework to investigate a topic beyond the scope of regularly scheduled classes. Projects may involve drone operations, sensors, data collection and analysis, equipment evaluation, or development of operational methods. The student and instructor establish project objectives, required work, progress meetings, and evaluation methods through an independent study agreement. Students document their work and present a final product, report, or demonstration appropriate to the project and units earned.

Credit and Hours

Variable hours Yes. Minimum units 0.5. Maximum units 2.0. All units are lab.
Lecture / lab / activity hours 0 / 1.5 to 6 / 0 curriculum hours per week. Semester totals are 0 lecture, 27 to 108 lab, and 0 activity hours. Outside work is 0 hours. Total student hours are 27 to 108.

Course Content

Fit within existing degrees or certificates No. No existing program is identified.
Course rationale Prepared students need opportunities to pursue substantial projects that extend beyond scheduled coursework. Independent study allows them to develop specialized skills, solve practical problems, and produce work that demonstrates their capabilities to employers. A flexible project format supports individual career interests and professional growth without replacing existing courses.
Transfer or articulation intent No.
Connection to the College Mission The course supports student success through individual learning goals and applied work connected to employment or advancement. Students develop initiative, technical judgment, and evidence of their skills for future professional opportunities.
Additional resources required _____
Resources and funding _____
Duplicate or overlap existing courses Yes. Projects build on AERO coursework and may connect with related technical disciplines. The course extends learning through individual projects and does not replace existing offerings. Project scope will be coordinated with relevant faculty. Consultation on the broader drone curriculum has established support from related disciplines.

Minimum Qualifications

Minimum qualification Aeronautics. Condition _____
Department currently has faculty meeting minimum qualifications _____

Course Cap

Tentative course cap category _____ Tentative course cap 25.
Effect on department efficiency A proposed cap of 25 retains the program's enrollment limit. Individual project agreements define the scope of work and instructor consultation needed for each student. Enrollment and project assignments must remain compatible with effective supervision.

Codes and Dates

Dean or director approval date _____
Originator Kyle Nessen. Co-contributor Ron Overacker.